

TIE TIY — rule_031 Compressed Audit Supplement

Consultant Briefing | Round 10 | Material change to deployment plan

Headline

Per Round 9 framework, we executed the compressed Phase 3 audit on rule_031 (7 audits: 5 mirroring rule_019's Phase 3 stack, 2 NEW designs specific to rule_031's sector specialization). All 7 audits completed.

Material finding: Audit 6 (sector substitutability — a NEW audit not derived from rule_019's chain) revealed that IT-sector specialization in rule_031 is **NOT load-bearing**. 9 of 12 non-IT sectors produce HIGHER walk-forward lift when substituted into rule_031's match logic. Specifically: Chem +29.3pp, Other +27.3pp, Auto +26.9pp, FMCG +22.5pp — all above rule_031's IT lift of 16.1pp. IT ranks 5 of 13 sectors for the Bear+UP_TRI+hot pattern.

Deployment recommendation reversed: Round 9's tentative "add rule_031 to shadow after compressed audit" → "do NOT deploy rule_031 as separate trade rule." Edge is real but is not the IT-specific edge the rule schema claims. It's a slice of a broader Bear+UP TRI+hot effect that rule_019 already captures more comprehensively.

Audit framework validation: Audits 1-5 (rule_019-pattern: label_randomization, date_shift, universe_perturbation, corp_action_spike, embargo_sensitivity) all produced PASS or AMBIGUOUS-favorable. Without Audit 6 — which Round 9 specifically called for — rule_031 would have entered shadow on the strength of those alone. Near-miss avoided.

1. Audit Cycle Results — 7/7 Complete

#	Audit	Verdict	Critical finding
1	label_randomization	PASS	z=2.44 vs scaled threshold (rule_019 used z=5.01 vs z>=3.0)
2	date_shift +1/-1	AMBIG/favor.	Both shifts collapsed below null band; +1 missed PASS by 0.05
3	universe_perturbation	PASS strong	Broad-based across 14 IT symbols; lift INCREASES at N=2,3
4	corp_action_spike	PASS clean	0 hard-threshold trades (vs rule_019's 15)
5	embargo_sensitivity	AMBIG/favor.	All 4 trials above null; step=120 +0.38pp drift (cleanest)
6	sector_concentration	FAIL	IT rank 5 of 13 sectors. NOT load-bearing. DECISIVE.
7	drift_vs_rule_019	Subset	rule_031 fires in 0 windows where rule_019 doesn't

Pre-Audit-6 read: rule_031 had a real, broad-based, robust edge (Audits 1-5). Recent-lift trajectory was IMPROVING (+9.5pp delta vs rule_019's -5.2pp). Looked like a favorable candidate for shadow.

Post-Audit-6 read: rule_031's edge is real but is a slice of a broader effect, not an IT specialization. The IT restriction reduces match volume by ~92% (2,275 → 189) while delivering mid-pack per-window lift. Adding rule_031 alongside rule_019 contributes operational complexity without diversification benefit.

2. Audit 6 — Sector Substitutability Detail

2.1 Design

NEW audit, not derived from rule_019's chain. Two parts:

Part A — IT-symbol concentration (ranked by match count): drop top-N IT symbols by count, re-run walkforward. Result: PASS. rule_031's edge survives top-3 IT symbol drops (broad-based within IT).

Part B — sector substitutability: for each of 12 non-IT sectors, construct a synthetic rule with IT replaced by that sector (regime=Bear, signal=UP_TRI, sub_regime=hot, sector=<control>). Run walkforward. Tabulate lift across all 13 sectors.

2.2 Sector substitutability results (all 13 sectors)

Rank	Sector	n_match_win	Lift	% pos windows	> shuffle_p95
1	Chem	18	+29.26pp	94.4%	Yes
2	Other	18	+27.28pp	88.9%	Yes
3	Auto	18	+26.88pp	83.3%	Yes
4	FMCG	21	+22.45pp	76.2%	Yes
5	IT (rule_031)	19	+16.07pp	78.9%	Yes
6	Pharma	17	+14.37pp	64.7%	Yes
7	Consumer	11	+13.84pp	72.7%	Yes
8	Infra	16	+13.37pp	75.0%	Yes
9	Metal	18	+12.86pp	66.7%	Yes
10	CapGoods	18	+10.88pp	66.7%	Yes
11	Bank	18	+8.20pp	61.1%	No
12	Energy	19	-1.23pp	52.6%	No
13	Health	11	-18.97pp	27.3%	No

2.3 Verdict mechanics

Both FAIL conditions met:

1. $IT_lift (16.07pp) \leq \max_other_sector_lift (29.26pp \text{ Chem})$. IT is not the strongest sector.
2. 9 non-IT sectors above shuffle_p95 (≥ 3 threshold). Pattern is broadly cross-sector, not IT-specific.

Ratio $IT/\max_other = 0.55x$. Threshold for PASS was $\geq 1.5x$.

3. Audit 7 — Drift vs rule_019

3.1 Window coverage (decisive)

	Count
Total walk-forward windows	182
rule_019 fires in	22
rule_031 fires in	19
Both fire in	19
rule_019 only	3
rule_031 only	0 (strict subset)

rule_031 fires in ZERO windows where rule_019 does not. Window coverage is strict subset. Adding rule_031 provides zero coverage extension at the window level.

3.2 Per-window lift correlation

Pearson $r = 0.158$ ($p = 0.520$, $n = 19$). Decision per spec: ORTHOGONAL ($r < 0.3$).

Caveat: at $n = 19$ with $p = 0.52$, the correlation is statistically indistinguishable from zero. True population r could be -0.3 to $+0.5$. Likely reflects sample-size noise from rule_031's small per-window match cohort (~18 matches per window vs rule_019's ~188), not genuine signal independence.

3.3 Sub-window comparison (interesting nuance)

Metric	Value
rule_019 lift in shared windows (19)	16.85pp
rule_019 lift in ALL its firing windows (22)	14.77pp
Delta	+2.07pp

When rule_031 fires, rule_019 averages 2pp HIGHER lift than its overall mean. rule_031's firing is a COINCIDENT INDICATOR of regime strength: when conditions are hot enough to produce IT matches (relatively narrow filter), the broader Bear+UP_TRI+hot regime is also producing strong cross-sector matches.

This is operationally interesting but does NOT justify deploying rule_031 as a separate trade rule. It raises a separate rule-design question: should rule_031 be repurposed as a confirmation/sizing overlay for rule_019? Surfaced as Open Question 1 below.

4. Strategic Implications

4.1 Deployment recommendation

rule_031: do NOT deploy as separate trade rule.

Reasons:

- Edge is real but NOT IT-specific (Audit 6: 9 of 12 non-IT sectors produce higher lift)
- Window coverage is strict subset of rule_019 (zero coverage extension)
- Per-window lift is noisier than rule_019 (smaller match cohort)
- Adds operational complexity without diversification benefit

4.2 Updated Bear-only sleeve composition

Pre-Audit-6 expected Bear-only sleeve: rule_019 + rule_031 (+ kill_001 filter). Post-Audit-6 actual sleeve: **rule_019 alone** (+ kill_001 filter).

rule_019 standalone characteristics (post-vp-fix): +14.77pp lift, 77% positive windows, fires in 22 of 182 windows (12.1% of historical record). Historical match volume 2,149 trades across 162+ symbols.

4.3 Discovery-replay implication (Path B)

Audit 6 surfaced 4 sectors (Chem, Other, Auto, FMCG) with stronger Bear+UP_TRI+hot edges than IT. None of these sector-specialized rules exist in the current rule library.

This is a **structural finding for the Path B discovery replay workstream**: did Lab discovery systematically underweight non-IT sectors? Possible causes:

- Leaky vp data biased sector cell boundaries during discovery (now fixed in canonical data)
- Selection biases in the rule generation pipeline favoring IT cohort
- Lab focused on specific sector slices that happened to include IT, not Chem/Auto/FMCG

Not acting on this finding now (no new rule design pending discovery replay results). Documenting for context. The four checks consultant outlined (cell-stability replay, selection-path replay, frontier-persistence test, freeze-and-reselect test) should specifically assess whether Chem/Auto/FMCG would have emerged as SURVIVORS under clean data.

5. Open Questions for Consultant

5.1 Question 1 — rule_031 as confirmation overlay?

Audit 7 surfaced a +2.07pp coincident-indicator delta: rule_019 averages 2pp higher lift in windows where rule_031 also fires. rule_031's firing is informative about WHEN rule_019 is unusually strong.

This raises a separate rule-design question, distinct from "deploy rule_031 as trade rule":

Should rule_031's firing be repurposed as a CONFIRMATION/SIZING overlay for rule_019? For example: "when both rule_019 and rule_031 fire, size rule_019 trade larger than when rule_019 fires alone."

If yes, this changes the shadow ops framework requirements (need to log rule_031 firing alongside rule_019 firing, even if rule_031 isn't generating trade cards). If no, rule_031 is fully retired.

5.2 Question 2 — pre-shadow workstream priorities

Round 9 specified four pre-shadow workstreams plus Path B in parallel. Given Bear-only deployment is now confirmed (rule_019 alone, no rule_031), should priority order change?

Workstream	Round 9 priority	Post-rule_031-cycle priority
Rarity-impact business audit	Required	Required (unchanged)
Regime-transition trust audit	Required	Required (unchanged)
Shadow ops framework	Required	Required + question 1 contingency
Compressed rule_031 audit	Required	COMPLETE — confirmed do-not-deploy
Path B discovery replay	Parallel	Parallel + Audit 6 finding

Implicit question: does the Bear-only constraint with rule_019 alone (vs rule_019 + rule_031) change the rarity-impact analysis? rule_019 alone fires in 22 of 182 windows. rule_019 + rule_031 would have fired in 22 + 0 = same 22 windows (since rule_031 is strict subset). So rarity is unchanged in window-frequency terms but signal-frequency within those windows is somewhat reduced (no rule_031 reinforcing fires).

5.3 Question 3 — discovery replay scope priority

Path B replay was scoped to: cell-stability, selection-path, frontier-persistence, freeze-and-reselect. Audit 6 added a specific question that wasn't in Round 9's scope: **did Lab discovery systematically underweight non-IT sectors?** Specifically, would Chem/Auto/FMCG sector-specialized Bear+UP_TRI+hot rules have emerged as SURVIVORS under clean (post-vp-fix) data?

Should this question be added to Path B scope, or treated as a separate post-replay follow-up workstream? Adding it to Path B might extend its scope; deferring it might delay discovery of stronger Bull/Choppy alternatives.

5.4 Question 4 — clarification on "income-capable" vs "valid sleeve"

Round 9 explicitly raised the underweighted business-model-mismatch concern: "Bear-only may be valid, yet still be the wrong production object to psychologically and economically anchor yourself to."

With rule_031 retired, rule_019 alone is the entire deployable system. ~12% historical firing rate. Now compounded by the structural finding that no non-Bear regime currently has deployable rules. The Bear-only constraint is sharper than before.

Specifically: should shadow deployment proceed with this narrower scope, or should we wait for Path B discovery replay to confirm/refute Chem/Auto/FMCG emergence before starting shadow ops? Tradeoff:

- **Shadow now (rule_019 alone):** faster path to operational validation, narrower edge

- **Wait for Path B:** potentially deploy multi-sector rule library (if Chem/Auto/FMCG emerge), 8-12 week additional delay

6. Repository State

All audit scripts, JSON outputs, and per-audit commits are version-controlled on branch `rule_031_audit` (off `vp-leakage-fix` at `e8395f95`). 7 commits total, one per audit. Available for review on request.

Awaiting consultant guidance on the four open questions before proceeding to next workstreams (rarity-impact business audit, regime-transition trust audit, shadow ops framework, Path B discovery replay).